

Non-Functional Requirements Document (NFRD)

1. Performance & Latency

NFR-1.1: Simulator predictions must resolve in less than 300 milliseconds under normal network conditions.

NFR-1.2: Global distribution charts must render in less than 500 milliseconds upon landing on the home page.

2. Reliability & Availability

NFR-2.1: The system must isolate database errors, falling back to clean data logs without server crashes.

NFR-2.2: Out-of-bounds input parameters must be validated and rejected with detailed error feedback.

3. Usability & Theme Toggling

NFR-3.1: The interface must transition smoothly between Light and Dark modes.

NFR-3.2: User theme choices must be remembered using localStorage across browser reloads.